



Clash Royale Winner Prediction

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Dataset collection

 BWANDOWANDO · UPDATED 2 YEARS AGO

▲ 81

<> Code

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Clash Royale S18 Ladder Datasets (37.9M matches)

Ladder Data of Season 18 Clash Royale

Data Card

Code (10)

Discussion (0)

Suggestions (0)

About Dataset

Context

I've been recently exploring **Microsoft Azure** and have been playing this game for the past 4 or so years. I am also a software developer by profession. I did a simple pipeline that gets data from the official **Clash Royale API** using (Python) Jupyter Notebooks and Azure VMs. I tried searching for public Clash Royale datasets, but the ones I saw don't quite have that much data from my perspective, so I decided to create one for the whole community.

I started pulling in the data at the beginning of the month of December until season 18 ended. This covers the season reset last December 07, and the latest balance changes last December 09. This dataset also contains ladder data for the new Legendary card Mother Witch.

The amount of data that I have, with the latest dataset, has ballooned to around **37.9 M distinct/ unique ladder matches** that were (pseudo) randomly being pulled from a pool of 300k+ clans. If you think that this is A LOT, **this could only be a percent of a percent** (even lower) of the real amount of ladder battle data. It still may not reflect the whole population, also, the majority of my data are matches between players of 4000 trophies or more.

I don't see any reason for me not to share this to the public as the data is now considerably large that working on it and producing insights will take more than just a few hours of "hobby" time to do.

Usability ⓘ
10.00

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Expected update frequency
Never

Tags

Arts and Entertainment

Tabular

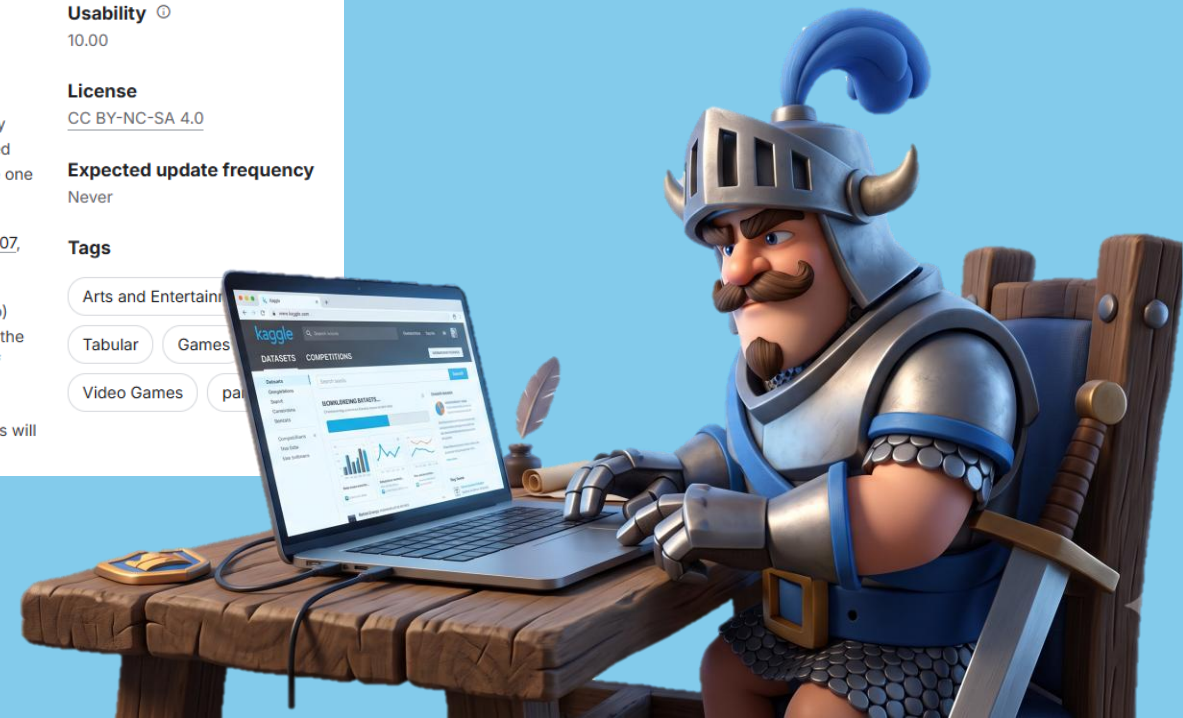
Video Games

Games

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<https://www.kaggle.com/datasets/bwandowando/clash-royale-season-18-dec-0320-dataset/data>



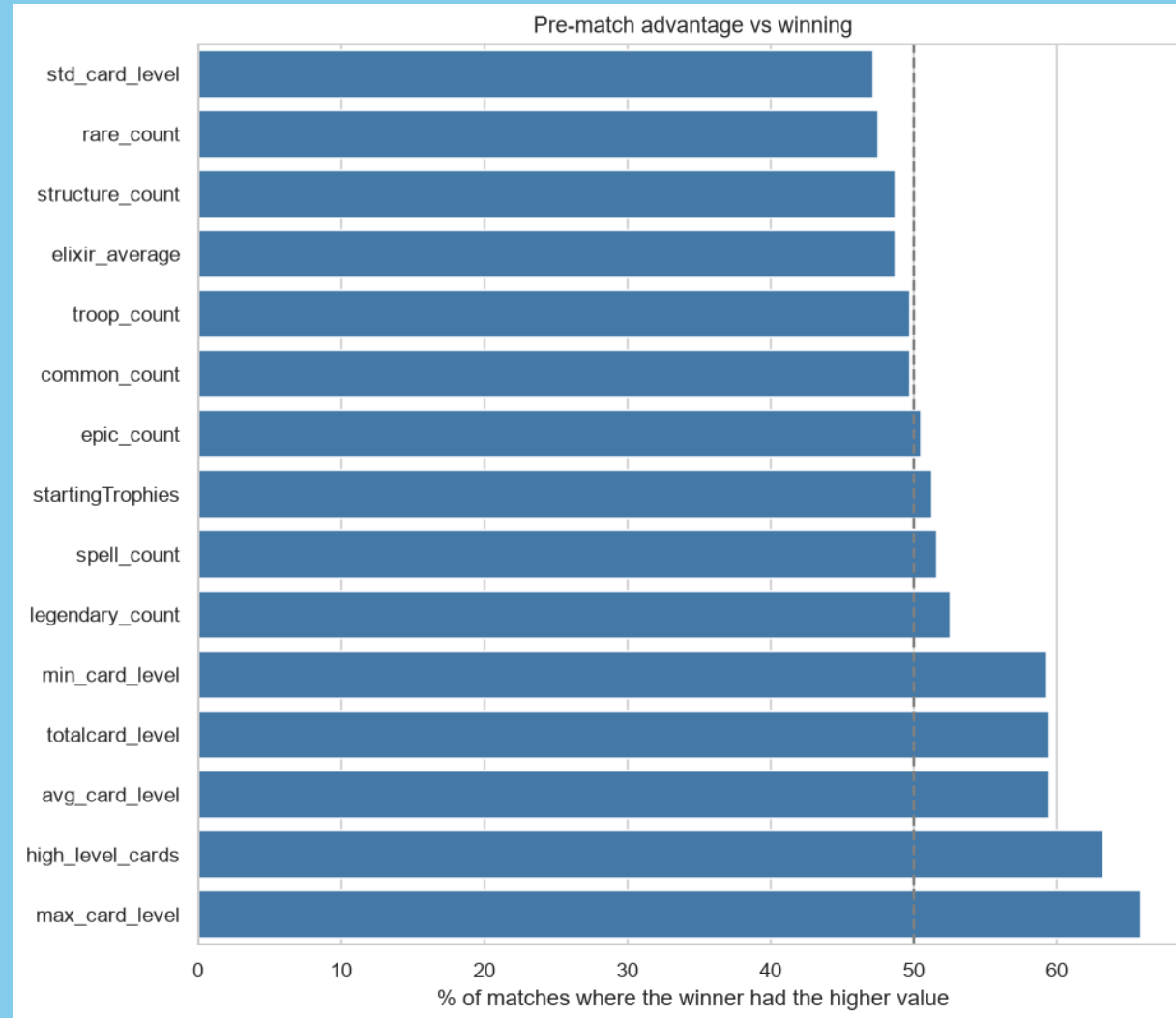
Who will win?



Arena
Rarity
Avg. Elixir
Trophies
Card Levels

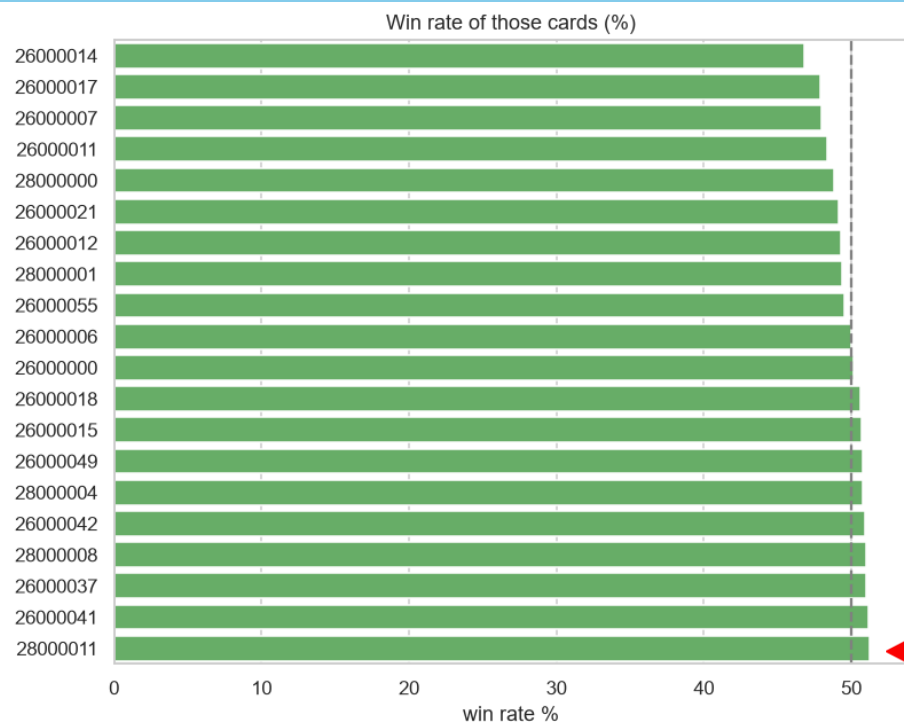
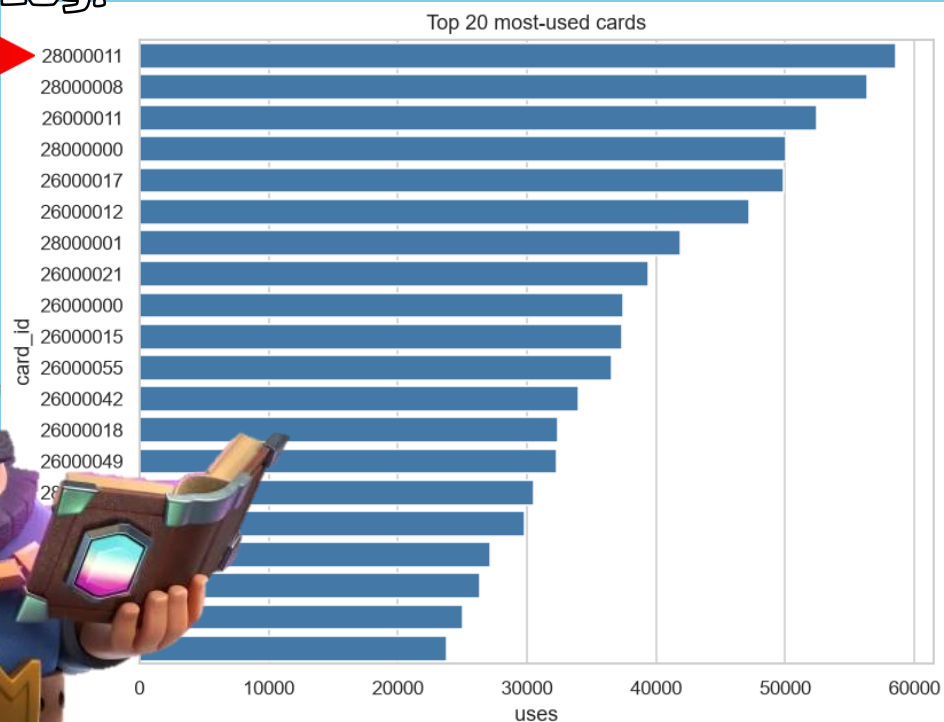
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Explanatory Data Analysis

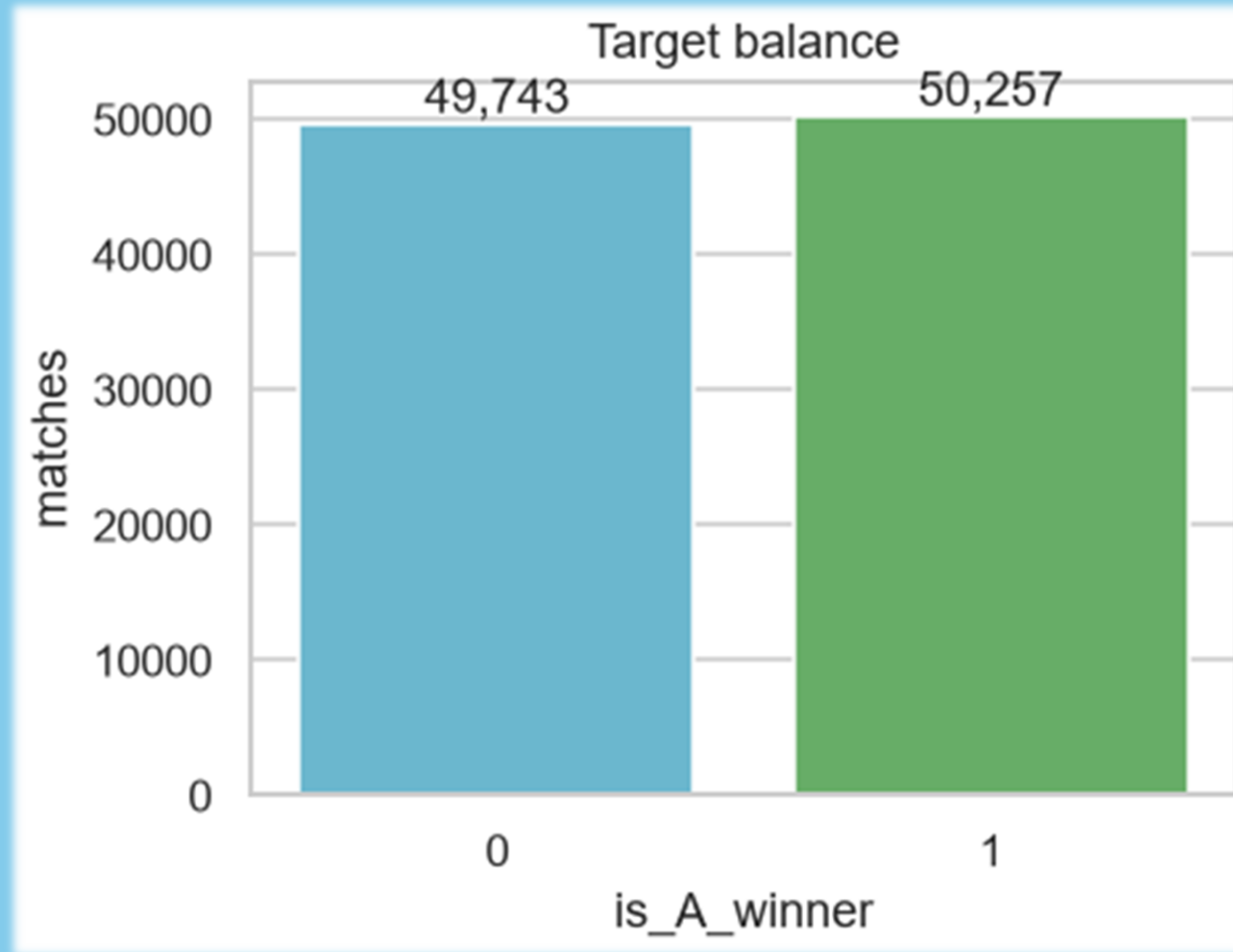


Explanatory Data Analysis

This is the Log!



Explanatory Data Analysis



Pipe line

load csv to Postgres

PREPROCESS

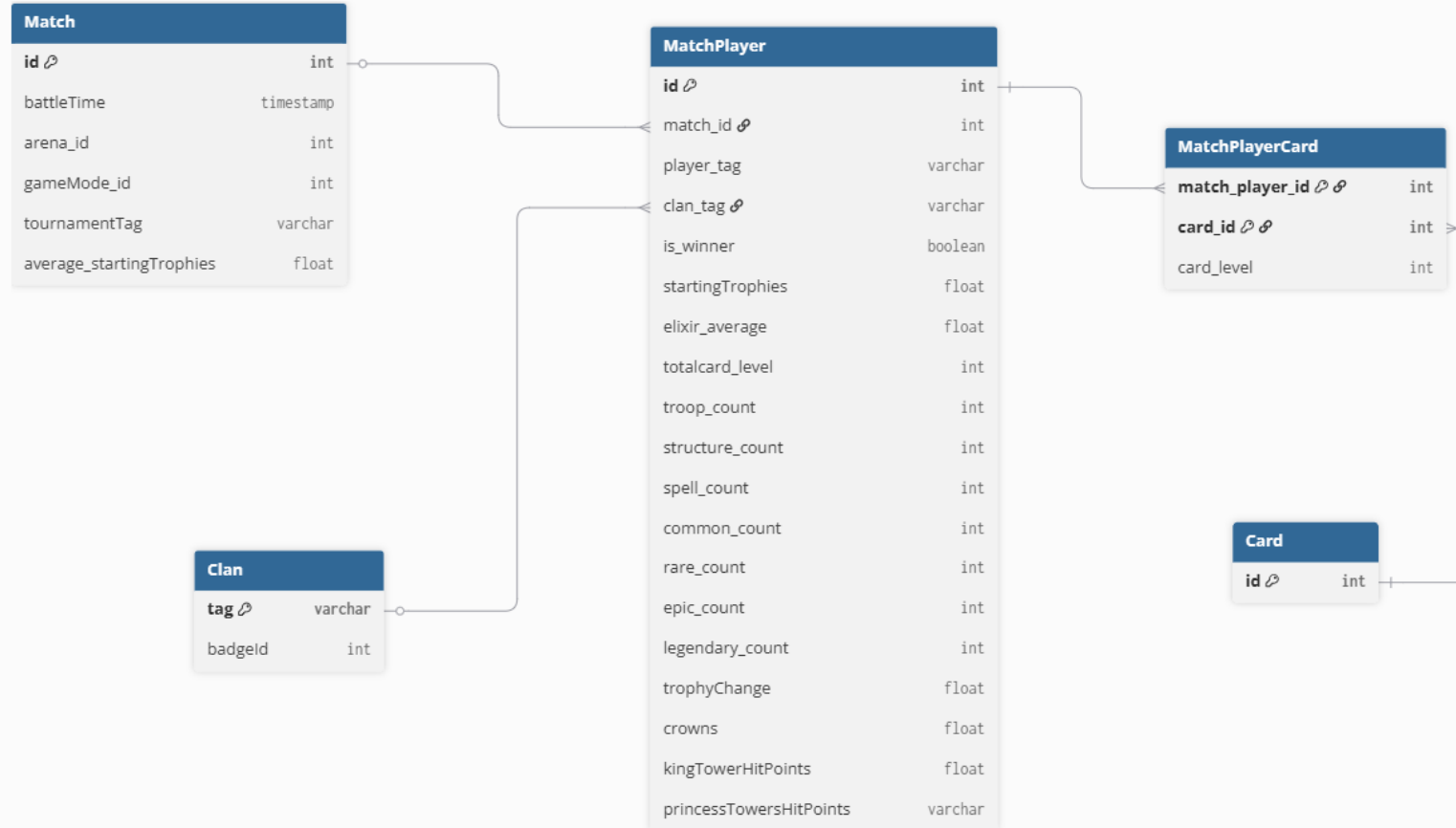
feature engineering

training

predictions & Evaluation



Loading Data to DataBase



Preprocess

Drop empty columns (tournament tag)

Missing Numeric Features : Median imputation

Missing Categorical features : Unknown imputation

Drop duplicates

Removing features causing data leakage



Feature Engineering



Comparison Features : $A_Col - B_Col$

Like : Average Elixir

Ratio Features : A_Col / B_Col

Like : Starting Trophy

Clan Tag Frequency : proportion of occurrences of clan tag in the training dataset

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Training

Logistic Regression

Random Forest

XGBoost

MLP



Predictions & Evaluation



Model	Accuracy	Precision	Recall	F1 Score	ROC AUC
XGBoost Top Performer	0.957750	0.952700	0.963901	0.958268	0.994329
Random Forest	0.956250	0.953902	0.959430	0.956658	0.991478
Neural Network	0.890917	0.878764	0.908594	0.893430	0.960591
Logistic Regression	0.604417	0.605109	0.615830	0.610423	0.671109



Predictions & Evaluation

Metrics for best model :

Accuracy Precision Recall F1 Score ROC AUC

0.955333 0.952467 0.959092 0.955768 0.993800

